

A Sequential Inspection and Replacement Policy for Degradation-based Systems

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SUMMARY

Condition-based maintenance (CBM) has been extensively studied. However, the majority of existing CBM research either consider a periodic inspection schedule or a fixed preventive maintenance threshold. While policies with periodic inspections and/or fixed maintenance threshold are easy to implement in practice, they may incur more-than-necessary inspections and induce more failures. In this paper, we develop a sequential CBM policy for systems subject to stochastic degradation. The aim of the proposed policy is to prevent or delay failures and perform maintenance activities just in time. Unlike conventional preventive maintenance that often fixes the inspection interval and the preventive maintenance threshold, both the next inspection time and the corresponding maintenance threshold in this paper are dynamically determined based on the current state of the system. The proposed sequential predictive maintenance policy is particularly important and applicable for general non-homogeneous degradation processes. The proposed model enables optimal scheduling of inspection and preventive maintenance decisions, in order to minimize the long-run maintenance cost rate including inspection, preventive and corrective maintenance costs. The performance of the proposed predictive maintenance policy is evaluated using a simulation-based optimization approach. Frequency of system failures and total maintenance cost rates are computed and compared with a bench mark maintenance policy, a periodic inspection/replacement policy. Our results show that there can be potential savings from the proposed predictive maintenance policy.

NOTATION

CR	corrective replacement
PR	preventive replacement
$X(t)$	cumulative degradation level at time t
α	shape parameter of Gamma process
β	scale parameter of Gamma process
$F(\cdot)$	CDF of Gamma process
$f(\cdot)$	PDF of Gamma process

$\delta(\cdot)$	inspection interval scheduling function
s	failure threshold
ξ	PR threshold
ξ_n	PR threshold at the n^{th} inspection
$\xi(\cdot)$	PR threshold prediction at time t
$C(t)$	total cost incurred in time interval $[0, t]$
c_{insp}	inspection cost
c_{pr}	PR cost
c_{cr}	CR cost
$N_{\text{insp}}(t)$	number of inspection actions in time interval $[0, t]$
$N_{\text{pr}}(t)$	number of PR actions in time interval $[0, t]$
$N_{\text{cr}}(t)$	number of CR actions in time interval $[0, t]$
γ	expected cost rate
u_{insp}	expected time duration of inspection
u_{pr}	expected time duration of PR
u_{cr}	expected time duration of CR
A	availability
N	number of replicates

1 INTRODUCTION

The importance of maintenance management is drawing more and more attention nowadays. Maintenance cost has kept increasing and become an important part of the production cost. For different industries, about 15% ~ 70% cost during production is associated with maintenance [1]. Various maintenance policies are developed to meet various situations, in order to reduce maintenance cost and increase the system reliability.

Maintenance strategies in general can be categorized as corrective maintenance (CM) and preventive maintenance (PM). PM can be further categorized as time-based preventive maintenance and condition-based preventive maintenance (CBM). Time-based preventive maintenance can lead to high maintenance cost and low availability due to the system condition is not taken into account. The maintenance decision of CBM is made based on the actual deterioration state [2].

The majority of CBM models either consider a periodic inspection schedule or a fixed preventive replacement threshold. One of the major problems of such a policy is that

more-than-necessary inspections may incur which leads to additional cost and potential damage to systems. Recently, sequential maintenance strategy becomes popular since prognostic information is used to obtain more accurate assessment of the current state and the future failure, and therefore make better decisions on inspection and/or maintenance schedules.

Grall *et al.* [3] propose a fundamental framework of predictive maintenance for systems subject to stochastic degradation. They take preventive maintenance (PM) threshold and inspection schedule as two decision variables for minimizing the long term cost rate. A mathematical model is developed using regenerative and semi-regenerative processes theory. Another approach for predictive maintenance decision making is to use the statistical information based on current system state [1]. Sensor-based information collection system is proposed and gives a good approach on how to capture the real-time system, and uses mean residual life distribution (RLD) as the stopping rule of the algorithm [4]. Lu *et al.* [5] give another criteria to decide what maintenance should be performed at an inspection. A cost model is built for comparison of the costs with and without performing PM. Deloux *et al.* [6] give a model for gradually deteriorating system, and optimize the maintenance policy by combining statistical process control and condition-based maintenance. Hashemian *et al.* [7] discuss three major predictive maintenance techniques based on different sources of data, which are existing sensor-based, test-sensor-based and test-signal-based technique. Machine learning methodology is presented for predictive maintenance [8]. Multiple classification modules with different prediction horizons are trained to provide tradeoffs in terms of frequency of unexpected breaks and unexploited lifetime, and then employing this information to minimize expected cost. Two approaches for k -out-of- n : F system are proposed. One is using the remaining useful lifetime of the components. The other one is applying the lifetime into the system and component [9]. However, the majority of the existing predictive models only consider a fixed PR threshold and/or a fixed inspection interval. As system deteriorates over time, it is more likely to fail, and it can be more beneficial to adjust the inspection interval and PR threshold to better prevent a system failure just in time.

In this paper, we propose a sequential CBM policy for systems subject to stochastic deterioration. The proposed model enables optimal scheduling of inspection and preventive maintenance decisions, in order to minimize the long-run maintenance cost rate including inspection, preventive and corrective maintenance costs. Unlike conventional preventive maintenance that often fixes the inspection interval and preventive maintenance threshold upfront, both the next inspection time and the corresponding maintenance threshold in this paper are sequentially determined based on the current state of the system. The proposed sequentially predictive maintenance policy is particularly important and applicable for capital-intensive industries (e.g. energy and defense) which require high

system availability and often incur high down time cost.

The remainder of the paper is organized as follows: Section 2 reviews basic concepts of the deterioration process and describes the proposed maintenance model. Section 3 provides numerical examples and compares the results from our proposed model to a periodic inspection/replacement CBM model. Section 4 concludes this study.

2 SEQUENTIAL CBM POLICY

2.1 Deterioration process

In this paper, we assume that the deterioration process can be described by a Gamma process. Gamma process is characterized by a shape parameter α and a scale parameter β . Deterioration increment of a Gamma process is not only always positive but monotonically increase, and is appropriate for describing failure processes of many devices. In this paper, we consider a homogeneous deterioration process. The deterioration between time t_1 and t_2 is only related to the time span.

Let $X(t)$ represent the degradation increment over time interval $[0, t]$. The CDF of $X(t)$ is given by

$$F(s; x_0, t) = \int_0^{s-x_0} \frac{1}{\Gamma(\alpha t) \beta^{\alpha t}} x^{\alpha t-1} \exp\left(-\frac{x}{\beta}\right) dx, \quad (1)$$

and the PDF of $X(t)$ is

$$f(s; x_0, t) = \frac{1}{\Gamma(\alpha t) \beta^{\alpha t}} x^{\alpha t-1} \exp\left(-\frac{x}{\beta}\right) \quad (2)$$

The mean and variance of the Gamma process are $\alpha\beta$ and $\alpha\beta^2$ respectively.

2.2 The Maintenance Model

In our proposed sequential CBM policy, the decision is made based on the current system state. More specifically, there are two decisions that have to be made at each inspection: (1) What is the time interval for next inspection? And (2) what is the PR threshold for next inspection?

We assume that the next inspection time interval is a function of the current system deterioration level, denoted as $\delta(X(t))$, where $X(t)$ is the degradation level at time t . For the PR threshold in next inspection, we denote the function as $\xi(X(t))$.

These two functions can take various function forms, for example, linear, step-wise linear, and exponential. In this paper, we assume that both functions take exponential functional forms. The relationship can be written as:

$$y = be^{ax} \quad (3)$$

Parameter a determines the decreasing or increasing rate of exponential function. Parameter b controls the first prediction value. In this paper, we assume a is smaller than 0, so inspection interval and PR threshold decrease as $X(t)$ increases. As the $X(t)$ increases, the system is more likely to fail. Thus, a smaller inspection interval and a lower PR threshold is desired to prevent the failure just in time.

We assume the failure of the system can only be revealed

through inspections. Let t_n denote the n^{th} inspection time since the last renewal. The proposed sequential CBM policy is described as follows.

- (1) Upon each inspection, if the degradation level $X(t_n) > s$, where s is the failure threshold, then CR is performed;
- (2) If $\xi(X(t_{n-1})) < X(t_n) < s$, PR is performed;
- (3) If $X(t_n) < \xi(X(t_{n-1}))$, do nothing.

The next inspection time interval and the next PR threshold are then computed based on the post-maintenance state of the system. If the system is correctively or preventively replaced, the degradation level will be reset to zero and next renewal cycle will start. If no maintenance action is performed, then calculate the next inspection time interval $\delta(X(t_n))$, and PR threshold $\xi(X(t_n))$.

The objective is to minimize the long run cost rate. The cost over interval $[0, t]$ is given by

$$C(t) = c_{\text{insp}} N_{\text{insp}}(t) + c_{\text{pr}} N_{\text{pr}}(t) + c_{\text{cr}} N_{\text{cr}}(t). \quad (4)$$

where c_{insp} , c_{pr} and c_{cr} are the cost of each inspection, PR and CR respectively. $N_{\text{insp}}(t)$, $N_{\text{pr}}(t)$ and $N_{\text{cr}}(t)$ are number of inspections, PR and CR actions during time $[0, t]$ respectively. Therefore, the long term cost rate is given by

$$\gamma = \lim_{t \rightarrow \infty} \frac{C(t)}{t} \quad (5)$$

Most existing analytical predictive maintenance models make various assumptions to simplify the real maintenance process, and a more flexible modelling approach is needed to provide closer-to-reality results. Therefore, we adopt a simulation approach for this problem. The proposed simulation model allows us to collect important performance statistics that are difficult to be obtained through analytical models.

2.3 Simulation model

The proposed simulation model is implemented in Matlab. Figure 1 provides the flow chart of a replicate. At the beginning of each replicate, we initialize the system by setting X_{now} , t_{now} equal to zero. X_{now} is the current degradation level, and t_{now} denotes the cumulative time elapsed in a replicate. Then, we calculate the next inspection time interval $\delta(X_{\text{now}})$, PR threshold $\xi(X_{\text{now}})$, and degradation increment $\Delta X(\delta)$. Then, we update t_{now} , t_{up} . The parameter t_{up} records the cumulative up time of the system in simulation. Upon each inspection, if the degradation level is below the PR threshold (ξ) for that inspection, we do nothing; if it is above the failure threshold s , then we perform CR. Otherwise, PR will be performed.

Figure 2 shows the simulation process of PR. We assume that the expected duration time of inspection, PR, CR and are constant, which are μ_{insp} , μ_{pr} , μ_{cr} respectively.

The procedure for CR is similar to this, only has different expected repair time duration.

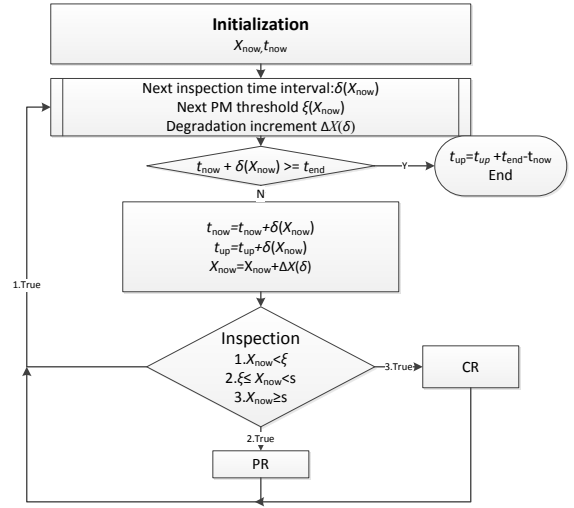


Figure 1. Flow chart of a replicate

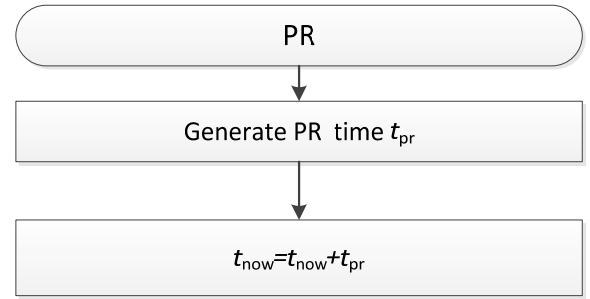


Figure 2. Flow chart of PR

The simulation results are used to calculate the average availability and cost rate, which can be expressed as follow:

$$A = \frac{\sum_{j=1}^N \frac{t_{\text{up},j}}{t_{\text{end}}}}{N} \quad (6)$$

$$\gamma = \frac{\sum_{j=1}^N \frac{C_j(t)}{t_{\text{end}}}}{N} = \frac{\sum_{j=1}^N \frac{c_{\text{insp}} N_{\text{insp},j}(t) + c_{\text{pr}} N_{\text{pr},j}(t) + c_{\text{cr}} N_{\text{cr},j}(t)}{t_{\text{end}}}}{N} \quad (7)$$

where N is the replicates number in simulation, and $t_{\text{up},j}$, $N_{\text{insp},j}(t)$, $N_{\text{pr},j}(t)$, $N_{\text{cr},j}(t)$ are t_{up} , $N_{\text{insp}}(t)$, $N_{\text{pr}}(t)$, $N_{\text{cr}}(t)$ of replicate j .

3 NUMERICAL EXAMPLE

In this section, we provide numerical examples to illustrate the proposed policy. Some baseline parameter values are summarized in Table 1. We use a_1 and b_1 to represent the parameters for inspection interval prediction, and a_2 and b_2 for the parameters of PR threshold determination.

First we examine the impact of the PR threshold. We fix a_1 to -0.005 and b_1 to 130. Figure 3 shows the exponential

model plot with different pairs of a_2 and b_2 .

Table 1. Fixed simulation parameters

Para-meters	Value	Para-meters	Value
N	1000	μ_{pr}	15
t_{end}	2000	μ_{cr}	100
s	400	c_{insp}	20
α	1	c_{pr}	200
β	2	c_{cr}	1500
μ_{insp}	2		

Table 2 illustrates the effect of b_2 when a_2 equals to -0.0001. When we have a small b_2 , the system will have more frequent PR actions, whereas a higher PR threshold leads to less PR actions. As the PR threshold increases as b_2 increases, there are more CR actions. Since CR is more expensive than PR, more failures often lead to higher cost rates.

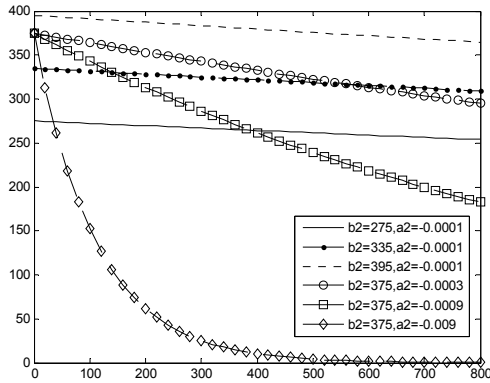


Figure 3. Plot of Maintenance Threshold as a Function of (a_2, b_2)

Table 2. Impact of b_2

PR threshold		number of inspection per replicate	number of CR per replicate	cost rate	availability
a_2	b_2				
-0.0001	255	16.819	0	1.334	0.896
	275	19.269	0	1.279	0.9
	295	20.657	0	1.25	0.902
	315	21.568	0.002	1.228	0.901
	335	23.801	0.027	1.225	0.903
	355	25.812	0.273	1.339	0.894
	375	26.535	1.331	1.959	0.854
	395	25.67	3.704	3.354	0.759
	415	8.613	8.163	6.208	0.579
	435	8.687	8.241	6.268	0.583

Table 3 shows the output change for a given b_2 . It's obvious that there is a smallest cost rate when $a_2 = -0.0007$. As the a_2 decreases, the PR threshold will decrease faster, which means that the chance of the renewal cycle ending up with PR will increase, thus the average CR actions will decrease and the cost rate will decrease.

Next, we examine the impact when a_1 and b_1 change. Figure 4 shows how inspection interval changes with different a_1 and b_1 .

Table 3. Impact of a_2

PR threshold		number of inspection per replicate	number of CR per replicate	cost rate	availability
a_2	b_2				
-0.0001	375	26.504	1.347	1.972	0.855
-0.0003		25.832	0.28	1.343	0.895
-0.0005		24.179	0.05	1.233	0.903
-0.0007		22.426	0.005	1.224	0.903
-0.0009		21.483	0.002	1.231	0.902
-0.001		21.298	0.002	1.235	0.901
-0.003		21.114	0.002	1.241	0.902
-0.005		21.081	0.003	1.241	0.902
-0.007		21.111	0.004	1.243	0.902
-0.009		21.131	0.002	1.241	0.902

We fix the a_2 to -0.0007 and b_2 to 335. Tables 4 and 5 summarize the parameter values used in the numerical experiments. From Table 4, cost rate achieves a lowest value when b_1 equals to 110. The reason is that the increase of b_1 can decrease the unnecessary inspections but will increase the probability that the system fails. Hence, it's definite that the trade-off between inspection and failure can come to optimal as b_1 increases.

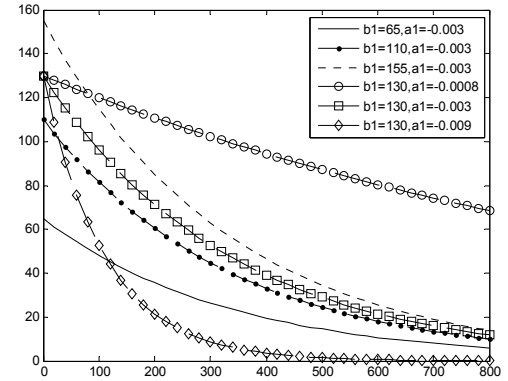


Figure 4. Plot of Inspection Intervals as a Function of (a_1, b_1)

Table 4. Impact of b_1

inspection interval		number of inspection per replicate	number of CR per replicate	cost rate	availability
a_1	b_1				
-0.003	50	50.488	0	1.583	0.869
	65	38.667	0	1.446	0.881
	80	31.364	0	1.334	0.893
	95	25.537	0.028	1.297	0.897
	110	21.071	0.025	1.241	0.901
	125	18.649	0.647	1.506	0.886
	140	14.983	3.518	3.155	0.781
	155	12.516	4.938	3.976	0.734
	170	10.991	3.191	2.921	0.794
	185	9.263	1.781	2.083	0.853

From table 5, the cost rate kept decreasing until $a_1 = -0.005$ and increased after. That's because the decrease of a_1 can shorten the inspection interval, which will lower the chance of the system failure. We can see that the CR per replicate is decreasing. However, one drawback is that this will result in too many inspections. Actually, when $a_1 = -0.011$, the inspections number is about 35 times, which is obviously unnecessary, and results in an increased cost rate.

Figure 5 depicts how inspection interval and PR threshold change for multiple renewal cycles within a replicate when $a_1 = -0.05$, $b_1 = 130$, $a_2 = -0.0001$, $b_2 = 395$.

Table 5. Impact of a_1

inspection interval	number of inspection per replicate	number of CM per replicate	cost rate	availability
a_1	b_1			
-0.0006	130	11.5	5.266	4.066
-0.0008		11.828	5.417	4.185
-0.001		12.008	5.503	4.255
-0.003		17.526	1.306	1.877
-0.005		21.134	0.004	1.242
-0.007		23.989	0	1.342
-0.009		28.432	0	1.429
-0.011		35.044	0	1.511

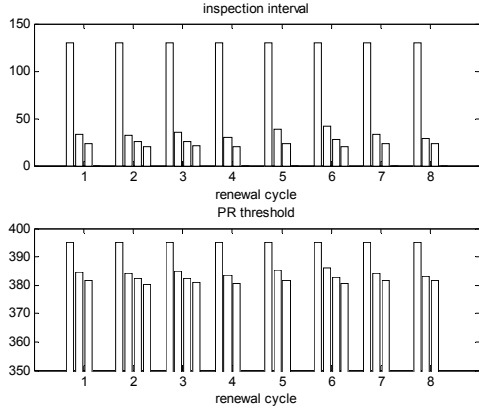


Figure 5. Change of inspection interval and PR threshold

Besides the impact analysis of model parameters, we offer a comparison of the proposed policy to a benchmark policy, wherein the inspection interval and PR threshold are fixed. The expected cost can be expressed as [10]:

$$\begin{aligned}
 C(\delta, \xi) &= \sum_{\lambda=1}^{\infty} (c_{pm} + \lambda c_i) \times \Pr(\text{PM at } \lambda \text{th inspection}) \\
 &+ \sum_{\lambda=1}^{\infty} (c_{cm} + \lambda c_i) \times \Pr(\text{CM at } \lambda \text{th inspection}) \\
 &= \sum_{\lambda=1}^{\infty} (c_{pm} + \lambda c_i) \times \int_0^{\xi} f(u; 0, (\lambda-1)\delta) [F(s-u; 0, \delta) - F(\xi-u; 0, \delta)] du \\
 &+ \sum_{\lambda=1}^{\infty} (c_{cm} + \lambda c_i) \times \int_0^{\xi} f(u; 0, (\lambda-1)\delta) [1 - F(s-u; 0, \delta)] du
 \end{aligned} \quad (8)$$

Rosenbrock's numerical search algorithm is applied to find out the optimal result of inspection interval δ and PR threshold ξ . The result shows that a minimum cost rate of 1.408 with availability of 0.884 are achieved when δ equals to 44.44 and ξ equals to 286.55. Obviously, most of the models proposed in table 2 to table 5 achieve a better cost rate and availability, which means potential cost savings can be obtained from the proposed sequential CBM policy.

4 CONCLUSION

This paper addresses an important question facing by reliability and maintenance managers – how to improve the system reliability/availability at reduced cost through advanced maintenance strategies. We propose a sequential CBM policy for inspection interval and PR threshold, which is considered as a better strategy than time-based maintenance. A simulation model is also developed to provide a general closer-to-reality framework for evaluating the performance of the predictive maintenance. Several

numerical examples are given to illustrate the effect of the model parameters, and prove that the proposed model has a better performance than a benchmark policy, in which inspection interval and PR threshold are considered fixed.

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